

Data Validation System

System Documentation

1. Overview

The Data Validation System is a desktop application built to collect basic personal information and check it against a set of rules before the record is saved. The form captures name, surname, ID number, age, email, contact number and gender. Each field is checked when the user presses Validate or Save, and the system displays a message box describing the problem if a field does not meet its rule.

The interface consists of a single window with labelled input fields on the left, radio buttons for gender, and a set of action buttons (Save, Help, Clear, Validate) on the right. The sections below describe the fields, the validation rules applied to each one, and the behaviour of the system in different scenarios, illustrated with screenshots taken from the running application.

2. Form Fields

The form contains the following fields: Name, Surname, ID No., Age, Email, Contact No., and Gender (Male or Female, selected using radio buttons). All fields are entered as free text except Gender, which uses a radio button selection.

3. Validation Rules

When the user presses Validate, the system checks each field in turn. If a field fails its rule, validation stops and a message box is shown describing what needs to be corrected. The table below summarises the rules currently applied.

Field	Rule Applied	Message Shown to User
Name	Must not be left empty	Name field cannot be empty
Surname	Must not be left empty	Surname field cannot be empty
ID Number	Must contain numbers only	Only use numbers for ID number field
Age	Must be a number above 19	Age should be above 19
Email	Must not be left empty	Email field cannot be empty
Contact Number	Must not be left empty	Contact number field cannot be empty
Gender	One option must be selected	Please select a gender

4. Buttons

Save — stores the entered information after it has passed validation, and confirms this with a success message. Validate — runs the validation checks on the current field values without saving. Clear — empties all fields on the form and confirms with a success message. Help — opens guidance for using the form.

5. Walkthrough

This section illustrates the system's behaviour using screenshots taken from the application, covering both invalid input and a successful submission.

5.1 Empty Name Field

If the Name field is left empty and the user presses Validate, the system stops and displays a message stating that the name field cannot be empty. The record is not saved until this is corrected.

The screenshot displays a web application titled "DATA VALIDATION SYSTEM". The form contains the following fields and values:

- NAME:** (empty text box)
- SURNAME:** Molaetsa
- ID NO. :** 1234567890987
- AGE:** 20
- EMAIL:** Kamo@gmail.com
- CONTACT NO. :** 0123456789
- GENDER:** ☒ MALE, ☐ FEMALE

A modal message box is displayed over the form with the text: "Name field cannot be empty". The message box has an information icon and an "OK" button. At the bottom right of the form, there is a pink box containing four buttons: "SAVE", "HELP", "CLEAR", and "VALIDATE".

Validation message shown when the Name field is left empty.

5.2 Invalid ID Number

The ID Number field only accepts numeric characters. If letters are entered, as shown below, the system rejects the entry and asks the user to use numbers only.

DATA VALIDATION SYSTEM

NAME: Kamohelo

SURNAME: Molaetsa

ID NO. : Kamohelo

AGE: 20

EMAIL: Kamo@gmail.com

CONTACT NO. : 0123456789

GENDER: ☒ MALE ☐ FEMALE

Message

Only use numbers for ID number field

OK

SAVE

HELP

CLEAR

VALIDATE

Validation message shown when the ID Number field contains non-numeric characters.

5.3 Age Below Minimum

The Age field must contain a value above 19. In the example below, an age of 17 was entered, which is below the minimum, so the system displays a message asking for an age above 19.

The screenshot shows a web application window titled "DATA VALIDATION SYSTEM". It contains a form with the following fields and values:

- NAME: Kamohelo
- SURNAME: Molaetsa
- ID NO. : 1234567890987
- AGE: 17
- EMAIL: Kamo@gmail.com
- CONTACT NO. : 0123456789
- GENDER: ☒ MALE ☐ FEMALE

On the right side of the form, there is a pink box containing four buttons: "SAVE", "HELP", "CLEAR", and "VALIDATE".

A "Message" dialog box is overlaid on the form, displaying an information icon (i) and the text "Age should be above 19". An "OK" button is located at the bottom right of the message box.

Validation message shown when the entered age is below the minimum allowed.

5.4 Successful Submission

Once all fields pass their validation rules, pressing Save stores the record and the system confirms this with a success message stating that the information has been captured.

DATA VALIDATION SYSTEM

NAME: Kamohelo

SURNAME: Molaetsa

ID NO. : 1234567890987

AGE: 20

EMAIL: Kamo@gmail.com

CONTACT NO. : 0123456789

GENDER: ☒ MALE ☐ FEMALE

SUCCESS

i Your information has been captured

OK

SAVE

HELP

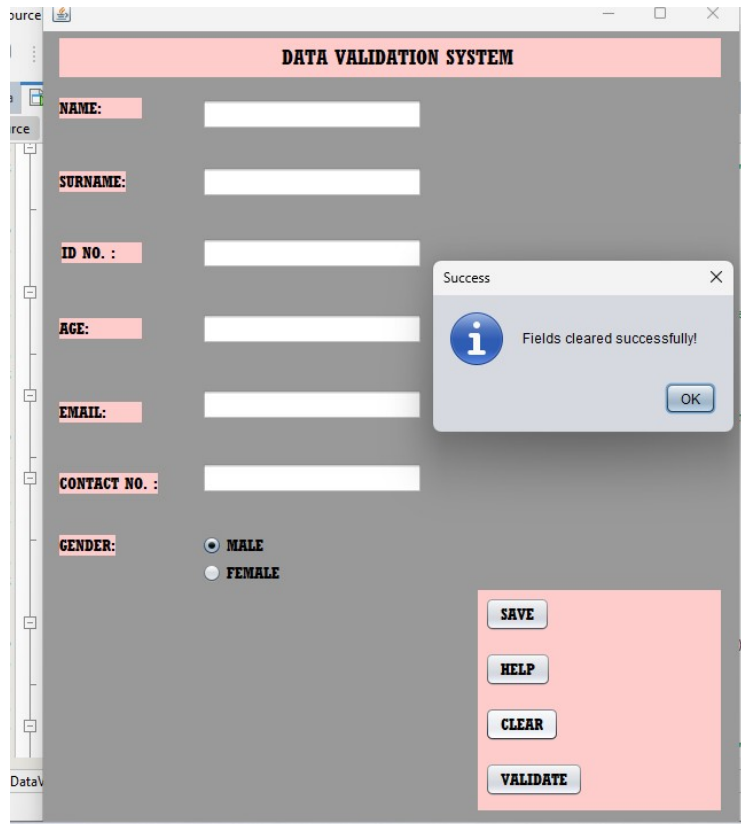
CLEAR

VALIDATE

Confirmation message shown after a record passes validation and is saved.

5.5 Clearing the Form

Pressing Clear empties all fields on the form, regardless of what was entered, and the system confirms this with a message stating that the fields were cleared successfully.



Confirmation message shown after the Clear button empties the form.

6. Summary

The system checks every field before allowing a record to be saved, rejecting empty required fields, non-numeric ID numbers, and ages below 19. Each rejection is paired with a specific message so the user knows what to correct. A record is only saved once all fields meet their rules, at which point the user receives a confirmation message.